VarudraForms

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Complete Project

From Zero to Live Launch — Everything You Need to Know

Daily operations forms platform for Indian businesses — in your language

Detail	Information
Product Name	VarudraForms (వరుద్ర ఫారమ్స్)
Company	AI TechWill — VarudraForms division
GitHub	github.com/VarudraForms/varudraforms
Email	varudraforms.in@gmail.com
Pilot Customer	Srinivasa Poultry Complex (SPC), West Godavari
Tech Stack	React + Firebase + Tailwind CSS
Target Launch	3 months from development start
Document Version	1.0 — April 2026

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1. What is VarudraForms?

VarudraForms is a daily operations management platform built specifically for Indian small and medium businesses. Think of it as a digital replacement for the paper registers, WhatsApp chaos, and Excel sheets that businesses currently use to manage their daily work.

1.1 The Problem We Solve

Every day, thousands of Indian businesses — poultry farms, rice mills, spinning mills, construction sites — record their daily data on paper. A poultry shed supervisor writes mortality counts, egg production, feed consumption in a register. His manager then types it all into an Excel sheet. The owner has no real-time visibility. If the register gets wet, or the Excel file gets corrupted, data is lost forever.

> 💡 *Real example: Our pilot customer Srinivasa Poultry Complex (SPC) manages 7 sheds with ~1,94,000 birds across poultry and spinning mill operations. They use 19 Excel sheets with complex formulas. This is their daily reality — and millions of Indian businesses face the same challenge.*

1.2 Our Solution

VarudraForms replaces paper registers and Excel sheets with a simple mobile app that:

- Works in Telugu (and Hindi/English) — uneducated workers can use it
- Works offline in sheds without internet — syncs automatically when connection returns
- Auto-calculates everything — opening birds, production %, feed per bird — no manual formulas
- Sends the owner a WhatsApp report every evening at 8pm
- Keeps Google Sheets updated automatically — teams don't have to change habits overnight
- Works on any Android phone (even ₹5,000 basic phones)

1.3 Who Is This For?

Business Type	What They Track	Pain Today
Poultry farm	Birds, mortality, eggs, feed, production%	Paper registers, Excel formulas
Spinning mill	Machine output, yarn production, shift report	Manual entry, no real-time data
Rice / grain mill	Paddy IN, rice bags, husk, godown stock	Stock mismatches, loss tracking
Construction site	Labour headcount, material, daily progress	Paper muster rolls, disputes
Wholesale shop	Opening stock, purchases, sales, cash	End-of-day manual calculations

> 💡 *Phase 1 builds for poultry (SPC as pilot). The same platform then serves all other business types — just a different form template.*

2. Project Overview

2.1 Key People

Role	Person	Responsibility
Product Owner / Developer	Ravi (AI TechWill)	Build the app, manage development, handle customers
Pilot Customer / Beta Tester	Friend (SPC Owner)	Use the app daily, provide feedback, pay after pilot
First Supervisor Users	7 SPC shed supervisors	Fill daily forms on phone every morning
Approver (SPC internal)	Krishna (SPC Manager)	Approve expense entries in the app
CA / Tax view user	SPC Chartered Accountant	View GST and financial reports only

2.2 Business Model

Plan	Workers	Price/Month	What's Included
Starter	Up to 100	₹999	Forms, checklists, reports, WhatsApp alerts, 1 business
Business	Up to 500	₹2,499	Everything + multi-business, supervisor logins, ECR export
Enterprise	Unlimited	₹4,999	Everything + bulk payments, worker WhatsApp bot, priority support
SPC (pilot)	~1,500	₹4,999	Enterprise plan — free for 3 months, then charged

2.3 Revenue Projections

Timeline	Customers	Revenue/Month	Status
Month 1-3	1 (SPC pilot)	₹0	Free pilot — validation phase
Month 4	1 paying	₹4,999	SPC starts paying after pilot success
Month 5-6	3-4 customers	₹12,000–₹18,000	2-3 new poultry farms in West Godavari
Month 9-12	10+ customers	₹45,000–₹60,000	Expand to mills, shops, other districts
Year 2	30+ customers	₹1,50,000+	Multi-industry, multi-language

3. Technology Architecture

3.1 Technology Stack — Plain English Explanation

Technology	What It Is	Why We Use It	Cost
React.js	Framework to build the web app interface	Sri already knows it. Fast, widely used.	Free
Vite	Tool that makes React development fast	Instant hot-reload — see changes immediately	Free
Tailwind CSS	Makes the app look good without writing CSS manually	Fast styling, consistent design	Free
Firebase Firestore	Google's database — stores all form data	Real-time sync, offline support, scales easily	Free tier → ₹800/mo
Firebase Auth	Handles login via phone OTP	Indian workers have phones, not emails	Free
Firebase Storage	Stores photos (expense bill photos)	Integrated with Firestore, secure	Free tier
Firebase Functions	Code that runs on Google's servers on a schedule	Sends WhatsApp report at 8pm daily	Free tier
Google Sheets API	Connects app to existing Google Sheets	Friend's team keeps using familiar Sheets	Free
Interakt (WhatsApp)	Sends WhatsApp messages from the app	Every Indian business runs on WhatsApp	₹999/mo
Netlify	Hosts the web app online	Free tier, auto-deploys from GitHub	Free

3.2 How Data Flows

Here is the complete journey of data from a supervisor's phone to the owner's WhatsApp:

1. Supervisor opens VarudraForms app on his Android phone at 8am
2. App shows pre-filled form: Opening birds = 21,704 (auto-loaded from yesterday's data)
3. Supervisor enters: Mortality = 4, Feed = 600kg, Eggs collected = 19,886
4. App instantly calculates: Production% = 91.8%, STD% = 89%, Difference = +2.8% (GREEN)
5. Supervisor taps Submit — data saved to phone's local storage immediately (works offline)
6. When internet available: data automatically syncs to Firebase Firestore (Google's servers, Mumbai)
7. All 7 supervisors submit their sheds throughout the morning
8. Every hour: Firebase Function reads Firestore and updates the Google Sheet automatically
9. At 8pm: Firebase Function reads all submissions and sends WhatsApp summary to owner
10. Owner opens WhatsApp — sees today's complete farm report without doing anything

3.3 Three Environments — DEV, QA, PRODUCTION

> ⚠️ *Never code directly in Production. Your client (SPC) uses Production only. You test everything in DEV and QA first.*

Environment	URL	Firebase Project	Who Uses It	Data
DEV (Development)	localhost:5173 (your Mac)	Firebase Emulator — free, local, no real data	Only Ravi	Dummy test data — reset any time
QA (Testing)	varudraforms-qa.netlify.app	varudraforms-uat Firebase project	Ravi + 1 tester	Sample SPC data — safe to share
PRODUCTION (Live)	app.varudraforms.in	varudraforms-prod Firebase project	SPC team daily	Real SPC data — never reset

4. Database Structure (Firestore)

Firestore is a NoSQL document database. Think of it like a very structured Google Sheet where every row is a 'document' and every column is a 'field'. Below is the complete structure for VarudraForms.

4.1 Collections Overview

Collection	What It Stores	Example Document ID
tenants	Each business that signs up for VarudraForms	spc-001
users	All user accounts with their roles	firebase_auth_uid_of_user
submissions	All daily form entries	spc-001_SPC3_2025-10-25
std_table	BV300 breed standard benchmarks (imported once)	bv_69.3
expenses	All expense entries (replaces PagarBook)	exp_spc001_20251025_001
sale_invoices	Egg sale invoices to NISHI and others	inv_spc001_20251025_006
tray_stock	Daily tray inventory movement	tray_spc001_20251025

4.2 Key Document Structures

tenants/{tenantId}

Field	Type	Example Value	Notes
name	string	Srinivasa Poultry Complex	Business name
type	string	poultry	Template: poultry | spinningMill | riceMill
plan	string	enterprise	starter | business | enterprise
sheds	array	["SPC1","SPC2","SPC3","SPC4","SPC5","SPC6","SPC7"]	List of locations
whatsappNumber	string	+919876543210	Owner's WhatsApp for reports
googleSheetId	string	1UVMvZNZWTbi...	Sheet ID from Google Sheets URL
createdAt	timestamp	2025-10-01	When they signed up

submissions/{tenantId}_{shedId}_{date}

Field	Type	Example	Entered or Auto?
date	string	2025-10-25	Auto — today's date

Field	Type	Example	Entered or Auto?
shedId	string	SPC3	Auto — from supervisor's login
tenantId	string	spc-001	Auto — from login
openBirds	number	21704	AUTO — previous day closing birds
mortality	number	4	ENTERED by supervisor
cullTransfer	number	0	ENTERED by supervisor
feedReceived	number	600	ENTERED — kg of feed
eggsCollected	number	19886	ENTERED by supervisor
damage	number	25	ENTERED — cracked/broken eggs
bigEggs	number	10	ENTERED — oversized eggs
proPercent	number	91.83	AUTO — (eggs/birds)*100
stdPercent	number	89.0	AUTO — lookup from std_table by age
diffPercent	number	2.83	AUTO — proPercent - stdPercent
feedPerBird	number	27.6	AUTO — (feed*1000)/birds in grams
eggBalance	number	1264	AUTO — yesterday balance + eggs - sales
birdAge	number	69.3	AUTO — weeks.days since placement
status	string	submitted	submitted | approved | rejected
submittedBy	string	uid_supervisor	Firebase Auth UID
submittedAt	timestamp	2025-10-25T08:34:00	Exact submit time

5. The Five Forms to Build

VarudraForms Phase 1 has exactly five data-entry forms. Everything else in the app is either auto-calculated dashboards or reports that read from these five forms.

Form	Maps to Excel Sheet	Who Fills It	Frequency
1. Daily Production Form	PRODUCTION REP	Shed supervisor (1 per shed)	Every day, per shed
2. Daily Checklist	daily check list	Shed supervisor	Every day, per shed
3. Expense Entry	DEEPAK RAJU CASH BOOK	Manager (Deepak/Raju)	As expenses happen
4. Sale Invoice	Sheet29	Office manager / owner	Per truckload of eggs
5. Tray Stock Entry	TRAY STOCK	Godown / store incharge	Daily

5.1 Form 1: Daily Production Form — The Most Important

This is the core of the entire application. Every other calculation depends on this form being filled correctly each morning.

Field Name	Telugu Label	Type	Auto or Enter?	Formula if Auto
Date	తేదీ	Date	Auto	Today's date
Shed No	షెడ్ నంబర్	Text	Auto	From supervisor's login
Bird Age	పక్షుల వయస్సు	Decimal	Auto	Days since placement ÷ 7
Opening Birds	ప్రారంభ పక్షులు	Number	AUTO	Yesterday open - mortality - cull
Mortality	చనిపోయిన పక్షులు	Number	ENTER	—
Cull/Transfer	తొలగించిన పక్షులు	Number	ENTER	—
Feed Received (kg)	వచ్చిన దాణా (కిలో)	Number	ENTER	—
Eggs Collected	సేకరించిన గుడ్లు	Number	ENTER	—
Box Sales	బాక్స్ అమ్మకాలు	Number	ENTER	—
Damage Eggs	పాడైన గుడ్లు	Number	ENTER	—
Big Eggs	పెద్ద గుడ్లు	Number	ENTER	—
Narration	నోట్స్ / వ్యాఖ్య	Text/Voice	ENTER	Voice input available
Production %	ఉత్పత్తి శాతం	Decimal	Auto	(Eggs ÷ Birds) × 100
STD Production %	ప్రమాణం %	Decimal	Auto	Lookup STD table by age

Field Name	Telugu Label	Type	Auto or Enter?	Formula if Auto
% Difference	తేడా %	Decimal	Auto	Actual% - STD% (Green/Red)
Egg Stock Balance	గుడ్ల నిల్వ	Number	Auto	Yesterday + Eggs - Sales
Feed per Bird	పక్షికి దాణా	Decimal	Auto	(Feed × 1000) ÷ Birds

5.2 Form 2: Daily Checklist — 47 Tasks

Every supervisor must complete this checklist daily for their shed. Each task is a simple YES/NO tap. The owner dashboard shows which tasks are incomplete.

Category	Tasks
Production & Admin	Production Report, Labour Payment, Labour Attendance
Equipment Check	ORP Meter, PF Meter, Water Bore SPC5, Water Bore SPC7
Feed & Transport	Feed Transport Report, Silo Cleaning (all 7 sheds)
Pest Control	Rat Kill, Fly Kill
Hygiene	Litter Spray, Sodium Hypochlorite, Side Mess Cleaning (all 7 sheds)
Health	Medicine Mixing (all 7 sheds), Vaccine (all 7 sheds)
Reports	Egg Stock Report, Egg Tray Report
Labour	Eggs to Labour, Birds to Lab

6. All Reverse Calculations (Excel → JavaScript)

'Reverse calculation' means the app automatically fills today's opening values from yesterday's data. The supervisor never types the same number twice. Below are all the formulas from the SPC Excel file, translated to JavaScript.

> ⚠️ *These formulas are the heart of the app. If any formula is wrong, all subsequent daily data will be wrong. Test every formula carefully against the existing Excel output before going live.*

Formula Name	Excel Logic	JavaScript Code	Example
Opening Birds	=PREV_OPEN - PREV_MORTALITY - PREV_CULL	openBirds = prev.openBirds - prev.mortality - prev.cullTransfer	55,960 - 109 - 0 = 55,851
Production %	=(EGGS/BIRDS)*100	proPercent = (eggs/birds)*100	(19886/21704)*100 = 91.6%
STD % Lookup	=VLOOKUP(AGE, STD_TABLE)	getStdByAge(age, stdTable).stdPercent	Age 69.3 → 89%
% Difference	=PRO% - STD%	diff = proPercent - stdPercent	91.6 - 89 = +2.6 (GREEN)
Egg Stock Balance	=PREV_BAL+PROD-SALE_BOX-SALE_TRAY-WASTE	eggBal = prev.eggBal + prod - (saleBox*210) - saleTray - waste	1000+19886-0-0-25= 20861
Tray Balance	=PREV+PURCHASED+NISHI_REC-USED	trayBal = prev.trayBal + purchased + nishiRec - used	1060+100+100-140= 1120
Feed per Bird	=(FEED_KG/BIRDS)*1000	feedPerBird = (feedKg/birds)*1000	(600/21704)*1000=27.6g
Feed per Egg	=FEED_KG/EGGS	feedPerEgg = feedKg/eggs	600/19886=0.030
Bird Age	=DAYS_SINCE_PLACEMENT/7	age = floor(days/7) + (days%7)/10	486 days → 69.4 weeks
STD Feed Required	=STD_FEED × BIRDS / 1000	stdFeedKg = (stdFeed * birds) / 1000	111g × 21704 / 1000 = 2409kg
TCS (sale invoice)	=MSG_AMOUNT × 0.001	tcs = msgAmount * 0.001	9,47,378 × 0.001 = ₹948
Total Eggs (invoice)	=BOXES × 210	totalEggs = boxes * 210	1152 × 210 = 241,920 eggs

> 💡 *The STD table has 2,105 rows — one for each week.day of the bird's life from week 1 to week 110. Import this entire table into Firestore once during setup. Never hard-code these values in the app.*

7. UX Design Rules — For Uneducated Users

VarudraForms must be usable by a 50-year-old shed supervisor who learned to use a phone only through WhatsApp. Every design decision must be tested against this person.

> ⚠️ *If a supervisor needs to be TAUGHT how to use the app, the design has FAILED. The app must be so obvious it requires zero training.*

Rule	What to Build	What to Avoid
Icons first, text second	Every button has a large emoji/icon above the label. Egg = 🥚, Mortality = skull icon, Feed = bag icon	Text-only buttons in English
Large tap targets	Every button minimum 56px tall. Full-width buttons preferred. No small checkboxes.	Tiny checkboxes, small radio buttons
Number pad for numbers	Show a large custom number pad (like a phone dial) for all numeric fields. Not a keyboard.	Regular keyboard for number entry
Pre-fill everything	Opening birds, date, shed number — all pre-filled. Supervisor enters only 5-7 values.	Empty forms that supervisor must fill from scratch
Traffic light colours	Green = good, Red = bad, Yellow = warning. No need to read numbers.	Black/white text output with no colour coding
Telugu by default	All labels, buttons, error messages in Telugu. English toggle in top bar.	English-only interface
Voice input for text	Notes/narration fields show microphone button. Supervisor speaks in Telugu.	Text keyboard for narration
WhatsApp-familiar patterns	Notifications look like WhatsApp messages. Share button shares to WhatsApp.	Email-style notifications
Max 3 taps to any task	Home screen shows today's pending tasks. One tap to open, fill, submit.	Deep navigation menus
Confirmation before submit	Show summary screen: 'You are submitting: Mortality 4, Eggs 19,886 — Correct?'	Immediate submit with no review

8. Development Roadmap — Week by Week

At 11.5 hours/week (weekday evenings 1-2 hours + weekend 4 hours), here is the complete plan from today to live launch.

Week	Phase	What to Build	Deliverable
Week 1	Setup	Fix Mac terminal, clone repo, create Vite React app, set up 2 Firebase projects (UAT + PROD), import STD table to Firestore, set Firebase budget alerts	localhost:5173 shows React app
Week 2	Auth	Phone OTP login screen in Telugu, Firebase Auth integration, user role system (owner/supervisor/auditor/ca), protected routes, Firebase Security Rules	Login works, roles redirect correctly
Week 3	Production Form Part 1	Daily production form UI — all fields with large number pad, Telugu labels, pre-fill opening birds from yesterday's Firestore, STD table lookup	Form opens with pre-filled data
Week 4	Production Form Part 2	All auto-calculations working live, validation (mortality < 2% alert), offline mode (Firebase persistence), submit to Firestore, traffic light for % diff	Complete form submits and saves
Week 5	Checklist + Supervisor Home	47-item daily checklist with large Y/N buttons, Supervisor Home screen showing today's tasks, pending items highlighted red	Supervisor can complete full daily workflow
Week 6	Owner Dashboard	Real-time dashboard: all 7 sheds status, total birds, today's mortality, production% vs STD%, egg stock, pending approvals. Matches DAY REP sheet exactly.	Owner sees live data from all sheds
Week 7	Expense Form	Expense entry form, photo capture for bills, category selection, submit for approval. Approval workflow: Deepak submits → Krishna approves → balance updated.	Replaces PagarBook completely
Week 8	Sale Invoice + Tray Stock	NISHI sale invoice form with auto-calc TCS/commission/final amount, payment tracking, tray stock daily entry form	All 5 forms complete
Week 9	WhatsApp Report + Sheets Sync	Firebase Cloud Function for 8pm daily WhatsApp report via Interakt API, hourly Google Sheets sync function, PWA manifest for Android install	Owner gets daily WhatsApp report
Week 10	UAT Testing	Deploy to QA branch → Netlify URL, run all 5 forms end-to-end with dummy SPC data, compare output vs existing Excel, fix any formula mismatches	QA sign-off — ready for pilot

Week	Phase	What to Build	Deliverable
Week 11-14	SPC Pilot — Parallel Run	Week 11: Both app + Excel running. Week 12: App primary, Excel mirror. Week 13: App only, Sheets read-only. Week 14: App is single source of truth.	Pilot complete — client trust built
Month 3+	Polish + Launch	Fix all pilot feedback, add spinning mill template, buy varudraforms.in domain, build marketing landing page, start charging ₹4,999/month	Live product with paying customer

9. Step-by-Step Coding Guide

This section tells you exactly what to code, in what order, and how to verify it works. Follow this sequentially — do not skip ahead.

9.1 Project File Structure

After setup, your project should look exactly like this. Every file serves a specific purpose.

File/Folder	Purpose	Build in Week
web/src/firebase.js	Firebase connection + emulator toggle + offline persistence	1
web/src/App.jsx	Routes + auth guard — redirects by role	2
web/src/pages/Login.jsx	Phone OTP login in Telugu	2
web/src/pages/OwnerDashboard.jsx	Real-time all-sheds dashboard	6
web/src/pages/SupervisorHome.jsx	Today's tasks for one shed supervisor	5
web/src/pages/AuditorView.jsx	Read-only report view for CA/Auditor	8
web/src/forms/ProductionForm.jsx	Daily shed entry form — the main form	3-4
web/src/forms/ChecklistForm.jsx	47-item daily task checklist	5
web/src/forms/ExpenseForm.jsx	Expense entry with photo, replaces PagarBook	7
web/src/forms/SaleInvoice.jsx	NISHI egg sale invoice with auto-calc	8
web/src/forms/TrayStock.jsx	Daily tray inventory entry	8
web/src/hooks/useFormulas.js	ALL reverse calculations as JavaScript functions	1 (write early)
web/src/hooks/useStdLookup.js	Fetch STD values from Firestore by bird age	3
web/src/hooks/useAuth.js	Current user + role from Firestore	2
web/src/hooks/usePreviousDay.js	Fetch yesterday's submission for pre-filling	3
web/src/hooks/useOfflineSync.js	Firebase offline persistence setup	4
web/src/components/NumberPad.jsx	Large number input — no keyboard	3
web/src/components/YesNoToggle.jsx	Big YES/NO buttons for checklist	5
web/src/components/TrafficLight.jsx	Green/Yellow/Red indicator	4

File/Folder	Purpose	Build in Week
web/src/components/OfflineBanner.jsx	Shows sync status	4
web/src/i18n/en.js	All English text labels	2
web/src/i18n/te.js	All Telugu text labels	2
web/src/templates/poultry.config.js	Form field definitions for poultry business	3
functions/dailyReport.js	8pm WhatsApp report Cloud Function	9
functions/sheetsSync.js	Hourly Google Sheets export	9
firestore.rules	Security rules — who can read/write what	2

9.2 Git Branch Workflow

Follow this every single coding session without exception:

11. git checkout develop && git pull origin develop (start of every session)
12. git checkout -b feature/what-you-are-building (create feature branch)
13. ... code, test on localhost ...
14. git add . && git commit -m 'feat: describe what you built' (save progress every 30 mins)
15. git push origin feature/your-branch-name
16. When feature is complete: git checkout develop && git merge feature/your-branch
17. When develop is stable: merge develop → qa for testing
18. When qa passes: merge qa → main for production

9.3 Commit Message Format

Prefix	When to Use	Example
feat:	Adding a new feature or form	feat: add production form with auto-calculations
fix:	Fixing a bug	fix: opening birds calculation off by one day
style:	UI changes — colours, sizes, layout	style: make number pad buttons larger for mobile
docs:	Updating documentation or README	docs: add STD table formulas reference
chore:	Non-code changes — config, dependencies	chore: update Firebase to v11
refactor:	Restructuring code without changing behaviour	refactor: extract formulas to useFormulas hook
test:	Adding tests	test: add formula unit tests

10. Testing Guide — How to Verify Everything Works

Testing is not optional. Every form must be tested with real SPC data before the pilot. If formulas produce wrong numbers, your friend loses trust and the pilot fails.

10.1 Formula Verification Testing

For each formula, test with real values from the SPC Excel file and compare outputs:

Formula	Test Input	Expected Output	How to Verify
Opening Birds	SPC3: 55,960 open, 109 mortality, 0 cull	55,851 next day opening	Check PRODUCTION REP row 21 in Excel
Production %	19,886 eggs, 21,704 birds	91.6%	Check PRO% column in PRODUCTION REP
STD % Lookup	Bird age 69.3 weeks	89%	Check STD sheet row for week 69
% Difference	91.6% actual, 89% STD	+2.6% (GREEN)	Check % DIFF column in PRODUCTION REP
Egg Stock	1000 balance + 19886 eggs - 0 sales - 25 waste	20,861	Check BALANCE STK column
Tray Balance	1060 open + 100 purchased + 100 received - 140 used	1,120	Check TRAY STOCK sheet
TCS on Invoice	₹9,47,378 MSG amount	₹948	Check TCS AMOUNT in Sheet29
Total Eggs (invoice)	1,152 boxes	2,41,920 eggs	Check TOTAL EGGS in Sheet29

10.2 User Role Testing Checklist

Test each role separately by logging in with a test account for that role:

Role	Can See	Cannot See	Can Do
Owner	All sheds, all financial data, all reports	Nothing hidden	Approve/reject all forms
Supervisor (SPC3)	Only SPC3 shed data	SPC1,2,4,5,6,7 data, financials, salaries	Fill SPC3 daily forms only
Auditor	Approved monthly reports, production totals	Raw daily entries, unapproved data, bank details	View and download reports only — no editing
CA / Tax View	GST invoices, sale records, PF/ESI reports	Operational data, bird counts, HR records	View and download tax reports only

10.3 Offline Mode Testing

19. Open app on phone — connected to WiFi, submit a form successfully
20. Turn off WiFi on the phone
21. Open app again — should load from cache (not show error)
22. Fill a new form and tap Submit
23. App should show: 'Saved locally — will sync when connected' yellow banner
24. Turn WiFi back on
25. App should auto-sync within 30 seconds — green tick appears
26. Check Firebase Console — the submitted form should now appear in Firestore

10.4 Device Testing Matrix

Device	Test What	Pass Criteria
iPhone (Safari)	Login, basic form fill, dashboard	No crashes, layout correct
Android ₹5,000 phone	Full workflow — login to submit	Works smoothly, number pad usable
Android ₹15,000 phone	Full workflow + offline mode	All features work
Windows laptop (Chrome)	Owner dashboard, reports, export	Tables load, PDF downloads
Mac (Safari)	Same as Windows laptop	No layout breaks in Safari

11. Data Collection — What to Get from SPC

Before writing any code for production, you need specific data from your friend. This data is used to set up the SPC tenant in Firestore and configure the app correctly.

11.1 Critical Data Needed BEFORE Coding Starts

Data Needed	Why Needed	How to Get It	Status
Chick placement date for each shed (SPC1-SPC7)	Drives bird age calculation — all STD lookups depend on this	Ask friend directly	Pending
Are SPC1-7 all poultry? Or some spinning mill?	Determines which template each shed uses	Ask friend	Pending
Who is 'Deepak' and who is 'Raju'? Same or different people?	Determines how many supervisor logins to create	Ask friend	Pending
Mobile numbers for all 7 shed supervisors	Firebase phone OTP login requires mobile numbers	Ask friend for list	Pending
Mobile number for 'Krishna' (expense approver)	Krishna gets approval notifications	Ask friend	Pending
Name of the spinning mill (if separate business)	Second tenant/template setup	Ask friend	Pending
Opening tray stock count (currently 1,060 trays per Excel)	Starting balance for tray inventory	Confirmed from Excel	Done ✓
Google Sheet edit access (for sync testing)	To test 2-way sync during pilot	Friend shares Sheet	Pending

11.2 Data to Import into Firestore Before Pilot

Dataset	Source	How to Import	Records
BV300 STD table	STD sheet in Excel file	Python script → JSON → Firestore batch import	2,105 rows
Shed master data	Sheet16 in Excel file	Manual entry in Firestore Console	7 sheds
Supervisor user accounts	Get from friend	Firebase Console → Auth → Add users manually	7 users
Opening stock balances	Latest Excel data	Enter as first submission with correct opening values	7 shed records

12. Transition Plan — Moving SPC from Excel to VarudraForms

The transition from Excel/PagarBook to VarudraForms must be done carefully. A sudden switch risks losing your friend's trust if anything goes wrong. This 4-week plan ensures zero data loss and zero disruption.

> ⚠️ *Never tell the team to stop using Excel until VarudraForms has been running correctly in parallel for at least 2 full weeks.*

Week	What Happens	Excel Status	App Status	Risk Level
Week 1	App running, supervisors fill BOTH app and Excel every day. Compare outputs daily. Fix any mismatches.	PRIMARY — official record	Secondary — testing only	Zero risk
Week 2	App is primary entry. Excel updated automatically via hourly sync. Team checks Sheet to verify sync.	Read-only — auto-updated by sync	Primary — all entries here	Low risk
Week 3	Team stops entering data in Sheet. App is the only entry point. Sheet is a report view.	Report view only — do not edit	Single source of truth	Very low risk
Week 4	Full confidence in app. Sheet kept open for reference. Start planning to charge from next month.	Archive / report view	Production system	No risk

12.1 Guarantee to Give Your Friend

Say this to your friend before the pilot begins:

> *"Anna, for the first 2 weeks we run both systems. If the app shows even one wrong number, we go back to the Sheet with zero data loss. Your Sheet data is always there as a backup. I will personally verify every number matches before asking you to trust the app completely."*

13. Pre-Launch Checklist — Everything Before Going Live

13.1 Technical Checklist

Item	How to Verify	Done?
All 5 forms work on mobile (Android + iPhone)	Test on real phones — not just browser	☐
Offline mode works (submit without internet, sync when online)	Turn off WiFi, submit, turn on — verify sync	☐
All reverse calculations match Excel output exactly	Compare 7 days of app output vs Excel	☐
Phone OTP login works for all 7 supervisor numbers	Test login for each supervisor phone	☐
Role-based access enforced (supervisor can't see other sheds)	Login as supervisor, try accessing other shed data	☐
Firebase Security Rules deployed to production	firebase deploy --only firestore:rules	☐
Budget alerts set at ₹500 and ₹2,000 on prod Firebase	Firebase Console → Billing	☐
PWA installs on Android from Chrome	Open app.varudraforms.in → Add to Home Screen	☐
WhatsApp daily report sent successfully (test)	Trigger function manually, verify message received	☐
Google Sheets sync working (app data appears in Sheet)	Submit form, wait 1 hour, check Sheet	☐
Firestore daily backup enabled	Firebase Console → Firestore → Backups	☐
SSL certificate active on domain	https:// shows padlock in browser	☐

13.2 Business Checklist

Item	Action Needed	Done?
varudraforms.in domain purchased	GoDaddy India → Buy .in domain → ₹499/year	☐
Domain pointed to Netlify	Netlify → Domains → Add domain → Update DNS	☐
Privacy policy page live	Required for WhatsApp Business API and App Store	☐
Terms of service page live	Basic terms document at varudraforms.in/terms	☐
Subscription agreement with SPC signed (informal ok)	WhatsApp message confirming ₹4,999/month from Month 4	☐

Item	Action Needed	Done?
Support channel set up	WhatsApp group: VarudraForms Support — Ravi + SPC team	☐
Feedback collection process	Weekly call with friend for first 4 weeks	☐

14. Project Success Metrics

How do you know VarudraForms is succeeding? Here are the measurable indicators at each stage.

14.1 Pilot Success (Month 1-3)

Metric	Target	How to Measure
Daily form submission rate	All 7 supervisors submit every day	Firebase Console → Firestore submissions count
Formula accuracy	App output matches Excel within 0.1%	Daily comparison of production% and egg balance
Offline reliability	Zero data loss when sheds have no internet	Check Firestore — all submissions eventually appear
WhatsApp report delivery	Owner receives report every day by 8:30pm	Owner confirms on WhatsApp group
Supervisor complaints	Zero complaints about app crashing	Support WhatsApp group — no messages = success
Sheets sync accuracy	Sheet auto-updated within 1 hour of submission	Compare Sheet timestamps vs submission timestamps

14.2 Launch Success (Month 4-6)

Metric	Target	Action if Not Met
SPC paying ₹4,999/month	First payment received by Month 4	Investigate why — offer extended free period
New customers signed up	2 more businesses by Month 5	Visit poultry farms in West Godavari personally
Monthly revenue	₹12,000+ by Month 6	Add spinning mill template to expand market
App rating from SPC team	Owner says 'much better than Excel'	Address top complaints from pilot feedback
Support requests per week	Less than 5 per week	Fix root causes — document FAQs

15. Complete Cost Summary

15.1 Development Cost (You Building Solo)

Item	One-Time Cost	Monthly Cost	Notes
Your development time	₹0 cash	—	138 hours over 12 weeks
Firebase (free tier during dev)	₹0	₹0	Use emulator locally
Firebase (after launch)	—	₹800–₹1,500	Only after free tier exceeded
Domain varudraforms.in	₹499/year	₹42/month	Buy from GoDaddy India
Netlify hosting	₹0	₹0	Free tier sufficient
Interakt WhatsApp API	₹0	₹999	Only needed for WhatsApp reports
Claude Pro (AI assistance)	—	₹1,700	Helps build 3x faster
Firebase emulator (local dev)	₹0	₹0	Completely free
TOTAL Phase 1 (3 months)	~₹3,000	~₹3,200/month	Roughly ₹12,600 all-in

15.2 Revenue vs Cost

Month	Revenue	Costs	Profit/Loss	Notes
Month 1-3	₹0	₹9,600	-₹9,600	Development + pilot (free)
Month 4	₹4,999	₹3,200	+₹1,799	SPC starts paying
Month 5-6	₹12,000	₹3,500	+₹8,500/mo	3 customers
Month 9-12	₹45,000	₹6,000	+₹39,000/mo	10 customers
Year 2	₹1,50,000+	₹20,000	+₹1,30,000/mo	30+ customers

> 💡 *Break-even happens in Month 4 — the first month SPC pays. From Month 5 onward, every new customer is almost pure profit since infrastructure costs are already covered.*

16. Quick Reference Card

Key Links and Contacts

Item	Detail
GitHub	github.com/VarudraForms/varudraforms
Email	varudraforms.in@gmail.com
Firebase Console	console.firebase.google.com
UAT App URL	varudraforms-qa.netlify.app (after Netlify setup)
Production URL	app.varudraforms.in (after domain purchase)
Netlify Dashboard	app.netlify.com
Interakt (WhatsApp)	app.interakt.ai
Dev environment	localhost:5173 (run: cd web && npm run dev)
Firebase Emulator UI	localhost:4000 (run: firebase emulators:start)

Daily Development Commands

Command	What It Does
cd ~/Desktop/varudraforms/web && npm run dev	Start React app on localhost:5173
firebase emulators:start	Start local Firebase (from repo root)
git checkout develop && git pull	Get latest code before starting work
git checkout -b feature/what-you-are-building	Create new branch for today's feature
git add . && git commit -m 'feat: description'	Save your work (do every 30 minutes)
git push origin feature/your-branch	Push to GitHub (backup to cloud)
npm run build -- --mode uat	Build for QA testing
firebase deploy --only hosting	Deploy to Netlify/Firebase hosting

If Something Goes Wrong

Problem	Likely Cause	Fix
App crashes on submit	Formula error — division by zero or null	Check useFormulas.js for null checks
Data not syncing to Sheet	Google Sheets API quota or auth expired	Check Firebase Function logs in Console

Problem	Likely Cause	Fix
WhatsApp report not sent	Interakt API key expired or quota hit	Check Interakt dashboard, renew if needed
Firebase bill spike	Infinite loop in code reading Firestore	Check Firebase budget alerts, check Function logs
Supervisor can't login	Phone number not in Firebase Auth	Add their number in Firebase Console → Auth
Opening birds wrong	Yesterday's submission missing or corrupted	Allow manual override of opening birds (add field)